Learning Modality Geometry for Multispectral Object Detection: Adaptive Spacing and 3-D Rotary Position Embedding

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***Abstract*—Positional encoding is crucial for the Transformer to effectively process multimodal feature information in multispectral object detection. However, existing studies often directly apply single-modal encoding designs to multimodal tasks, resulting in positional encoding focusing on spatial information while neglecting modality-specific information, which limits the performance of the Transformer. To explore the potential of positional encoding in jointly processing spatial and modality information in the multimodal domain, we propose an adaptive multimodal rotary position embedding transformer (AMRoFormer) for multispectral remote sensing object detection. It efficiently exploits multimodal spatial and modality information. The AMRo-Former consists of the multimodal rotary position embedding cross attention (MRoCA) and adaptive intermodal spacing module (AISM), where MRoCA includes multimodal rotary positional embedding (MRoPE) and cross-attention (CA) mechanism. We introduce a 3-D positional encoding strategy that addresses the challenges associated with 2-D positional encoding, particularly the difficulty of retaining modality-specific information. We design MRoPE by applying the 3-D positional encoding strategy and rotary position embedding (RoPE), which enables the independent processing of modality information through separate 2-D encoding and mixed 3-D encoding. Based on MRoPE, we further introduce MRoCA, the first multimodal attention mechanism to integrate a 3-D positional encoding strategy and to introduce the concept of intermodal spacing. To better exploit intermodal spacing while reducing the costs associated with manually setting, we design a lightweight AISM. Integrated into a dual-stream backbone network, our method achieved state-of-the-art (SOTA) performance in multispectral object detection, attaining the highest detection evaluation metrics to date on the vehicle detection in aerial imagery (VEDAI), drone-vehicle, and low-light vision infrared-visible paired (LLVIP) datasets, with mAP50 scores of 0.8962, 0.8531, and 0.9937. Overall, AMRo-Former demonstrates exceptional generalization performance in multispectral object detection.**

***Index Terms*—Attention mechanism, feature fusion, multimodal, multispectral object detection, position encoding (PE), transformer.**

I. Introduction

In RECENT years, optical imaging technology has witnessed remarkable advancements [1], [2], significantly driving progress in remote sensing object detection research. This technology has extensive applications across civilian and military domains, including vehicle monitoring, urban traffic management, and emergency rescue missions. However, practical implementations still face challenges, including environmental occlusions and adverse illumination conditions [3]. Under such challenging scenarios, relying solely on single-modality information becomes inadequate to satisfy the increasingly stringent accuracy requirements for remote sensing object detection. Therefore, the integration of the advantages of different modalities, specifically by leveraging multispectral images which combine the rich textural details and color details of visible spectrum imagery and the thermal radiation information from infrared images [4], [5], has emerged as a critical research focus.

With the development of dual-stream backbone networks, recent works have adopted CNN-Transformer architectures for multispectral object detection [6], [7], capitalizing on the complementary strengths of convolutional neural networks (CNNs) [8], [9], [10] and Transformers [11], [12], [13]. CNNs excel at extracting fine-grained local features within individual modalities, while Transformers, through self-attention mechanisms, are adept at capturing long-range dependencies and facilitating cross-modal feature interactions. To exploit these respective advantages, many existing works explicitly assign distinct roles to each component [14], [15]: CNNs are typically responsible for intramodality feature extraction, whereas the Transformer focuses on cross-modality fusion and semantic enhancement. This division of labor is typically operationalized via fixed architectural patterns or concatenation strategies.

Unlike the implicit positional awareness of CNNs, which inherently encode positional information through local receptive fields and layer-wise propagation [16], Transformer modules require the explicit introduction of the position encoding (PE) method to incorporate position information. The design of the PE strategy is a critical determinant of the model's performance. Current research predominantly emphasizes innovations in PE methods, including data-adaptive positional encoding [17], conditional positional encoding [16], and rotary position embedding (RoPE) [18]. In image

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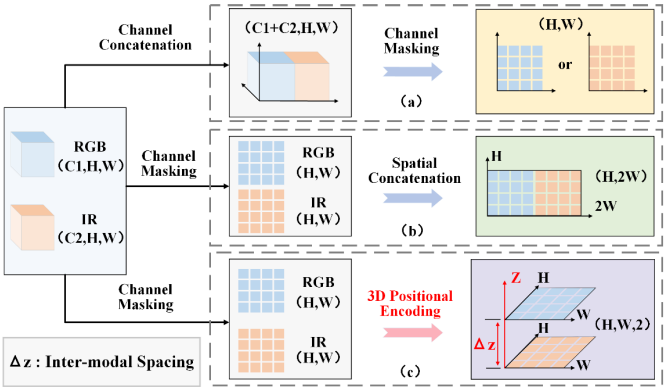


Fig. 1. (a) and (b) Channel concatenation and spatial concatenation strategies. (c) Our 3-D positional embedding strategy, where an orthogonal axis (Z) is separated from the spatial plane to represent modality position information.

processing tasks, a 2-D position embedding strategy preserves the inherent 2-D spatial relationships among input tokens, thereby achieving superior performance compared to 1-D methods [19], [20]. For multimodal tasks, most methods employ a 2-D positional encoding strategy using either channel concatenation [21], [22] or spatial concatenation [23], [24], reshaping multimodal features into 2-D formats compatible with traditional unimodal processing frameworks. From a geometric viewpoint, both channel concatenation and spatial concatenation can be interpreted as flattening an underlying 3-D positional space (two spatial dimensions plus one modality dimension; see Fig. 1). However, PE after channel concatenation [Fig. 1(a)] is typically applied solely in the spatial domain, ignoring modality-specific information. This blurs the intrinsic distinctions between modalities, failing to quantify the positional relationships between cross-modal tokens. Conversely, spatial concatenation [Fig. 1(b)] spatially separates modality-specific features but inevitably conflates or distorts the spatial alignment between modalities, leading to compromised cross-modal consistency and reduced fusion effectiveness. Beyond these flattening schemes, recent studies have started to design PE strategies tailored to higher dimensional scenarios. One line of work tackles 3-D positional information by learning projections from 3-D space to a lower dimensional token space [25], [26]. However, the extra geometric dimensions are typically compressed into feature channels, rather than modeled as explicit axes/dimensions, and they are not intended to represent cross-modal relationships. More recently, researchers have begun to emphasize explicit modeling of higher dimensional token spaces: depth, camera pose, and other geometric attributes can be injected as positional cues to improve structural awareness [27], [28]. Nevertheless, these strategies are usually designed for specific scenarios and do not scale gracefully when multiple axes must be handled jointly. In summary, existing strategies should be viewed as compromises made in the absence of an effective cross-modal positional encoding scheme, forcing the model to allocate more of its representational capacity to compensate for the missing positional information.

In addition, current positional encoding paradigms face limitations in processing multispectral remote sensing images. Present implementations of 2-D positional encoding in image processing transformations primarily utilize absolute positional encoding (APE) [11] and relative positional encoding (RPE) [19], [33]. Compared to APE, RPE provides enhanced local context perception beneficial for object detection but incurs substantial computational overhead and limited extrapolation capability for high-resolution imagery. The RoPE [18] method achieves RPE via complex-domain rotational transformations, offering low computational overhead and superior generalization capabilities for long-range extrapolation, which makes it suitable for high-resolution remote sensing object detection [20]. Multimodal RoPE variants have recently emerged as a promising direction [29], [30], [31]. They extend RoPE to multiple axes by allocating frequency bands across dimensions, enabling unified modeling for multiaxis layouts [32]. However, these approaches are mainly designed around spatial–temporal or vision–language structure and treat visual tokens from different modalities as a single stream. There is no explicit modality-identity that can be jointly encoded with spatial position, nor any learnable notion of intermodal spacing. This gap motivates our 3-D multimodal rotary positional embedding, where the third axis is devoted to the modality dimension and is further modulated by a learnable intermodal spacing module, see Fig. 1(c).

In our work, we develop a 3-D multimodal position embedding strategy to embed modality-specific information and spatial positional information within a unified position embedding space, enabling quantifiable distances for cross-modal tokens while maintaining spatial alignment information. Specifically, as shown in Fig. 1(c), we design a 2-D xy plane to represent spatial position. We introduce a third, orthogonal z-axis that encodes modality identity, yielding distinct modality planes for dual-modality data. Consequently, the relative positions of cross-modal tokens become explicitly quantifiable in this 3-D embedding space. Crucially, we introduce the concept of intermodal spacing Δz generated by the relative positions of cross-modal tokens. Intuitively, the magnitude of Δz implicitly influences the strength of cross-modal interactions, and different datasets or feature distributions may have different optimal values of intermodal spacing for best performance. Therefore, there exists an optimal Δz for model performance. To set Δz, two implementation routes are conceivable: 1) a fixed scalar value which may be obtained from exhaustive grid search; and 2) a learnable scalar produced by a small auxiliary regression network, which can be seamlessly integrated into end-to-end training. We adopt the latter as the implementation for our method, and we will compare the impact of these two approaches in Section V.

Building upon this, we propose a novel Transformer architecture for multispectral object detection, termed the adaptive multimodal RoPE transformer (AMRo-Former), which integrates our proposed 3-D multimodal rotary positional embedding strategy. Specifically, we introduce multimodal rotary positional embedding (MRoPE) to implement this strategy, extending 2-D-RoPE by incorporating mixed-axis frequency encodings to differentiate modalities. MRoPE is integrated into the cross-attention (CA) mechanism, yielding a multimodal rotary position embedding cross attention (MRoCA) module.

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To optimize the intermodal spacing (Δz), we further propose a lightweight adaptive intermodal spacing module (AISM), which automatically learns optimal spacing configurations, thus improving performance with minimal additional computational costs. AISM and MRoCA fuse multimodal features at multiple hierarchical levels within a dual-stream backbone, together forming the AMRo-Former architecture. The main contributions of this article are listed as follows.

1) We propose a novel 3-D multimodal rotary positional embedding strategy that explicitly defines a unified 3-D positional embedding space to encode both spatial and modality information. This design enables explicit quantification of distances between cross-modal tokens while preserving modality-specific characteristics and spatial alignment. Based on this strategy, we develop MRoPE using mixed-axis frequencies. MRoPE is then integrated into a MRoCA module for cross-modal feature fusion.
2) Building upon the 3-D multimodal rotary positional embedding strategy, we propose the critical concept of intermodal spacing, which defines the distance between modalities and enables the quantification of distances between cross-modal elements. Furthermore, we design a lightweight AISM that automatically determines the optimal intermodal spacing by analyzing multimodal input features. This module eliminates the need for manual tuning, ensures minimal computational overhead, and, in turn, allows the model to achieve improved performance.
3) By integrating AISM and MRoCA, we develop the AMRo-Former architecture and embed it into a dual-stream backbone network for validation. Extensive experiments on multiple public datasets demonstrate that our method outperforms baseline and state-of-the-art (SOTA) approaches. Specifically, our model achieves mAP50, mAP75, and mean average precision (mAP) scores of 0.8962, 0.7622, and 0.6064, respectively, on the vehicle detection in aerial imagery (VEDAI) dataset; and mAP50, mAP75, and mAP of 0.8531, 0.7730, and 0.6362, respectively, on the drone-vehicle dataset; and mAP50, mAP75, and mAP of 0.9937, 0.8984, and 0.7641, respectively, on the low-light vision infrared-visible paired (LLVIP) dataset, respectively, surpassing all existing the SOTA results.

II. Related Work

A. Positional Encoding for Attention Mechanism

The self-attention mechanism is permutation invariant, which necessitates the use of positional embedding design to incorporate sequence order. APE [11] provides a unique positional representation for each token in the sequence directly related to its position. In the vanilla Transformer, this is achieved by encoding sinusoidal functions of varying frequencies, which are added to the input, or by using learnable parameters to embed positional information more effectively into the self-attention mechanism. RPE [33], by focusing on the relative distances between tokens rather than their absolute positions, enables the model to capture position-invariant relationships. This means that even if the sequence is shifted, the model can still capture relationships between tokens. Building on RPE, RoPE [18] integrates relative positional information directly into the attention mechanism via complex-valued rotations, enabling efficient and theoretically sound modeling of token pairwise positions without introducing additional parameters. Recent research has demonstrated the effectiveness of RoPE in enhancing large language models [34] and improving their ability to handle long texts, and it has been integrated into the vision transformer architecture [20], which has shown significant improvements in model performance for object detection tasks. Standard APE, RPE, and RoPE were originally developed for 1-D sequences and thus do not natively capture the structural priors of higher dimensional signals. Although a number of studies have extended 1-D encodings to 2-D for image applications, such extensions are often realized by introducing additional spatial axes via direct factorization or axial decomposition, because the added dimensions remain homogeneous to the original one. Representative examples include iRPE, which develops image-dedicated 2-D relative encodings with directional modeling [19], and RoPE-ViT [20], which systematically analyzes how to instantiate RoPE on 2-D lattices and proposes mixed-axis designs to better reflect 2-D spatial geometry and extrapolate to higher resolutions.

However, higher dimensional information is often heterogeneous to the 2-D image plane, e.g., modality identity, sensor geometry, and acquisition-related attributes, whose semantics are fundamentally different from Euclidean spatial offsets. In such cases, a naive axis-wise extension of 1-D/2-D PE is insufficient to represent the extra factors in a geometrically meaningful way. To inject heterogeneous cues into token interactions, several 3-D perception models work map high-dimensional geometry into a shared embedding space. PETR [25] encodes 3-D coordinates into image features via position embedding transformation to obtain 3-D position-aware tokens, while 3DPPE [26] further incorporates depth priors to approximate 3-D point locations for multicamera detection. Meanwhile, geometry-aware attention mechanisms emphasize explicit relative modeling at the attention level. GTA [27] encodes token relations as relative geometric transformations derived from the query–key geometry, and Cameras-as-RPE [28] studies camera-conditioned transformers and formulates camera intrinsic and extrinsic as relative positional information. These methods collectively validate the benefit of injecting extra position information (e.g., depth/camera geometry) into attention, but the additional factors are typically mixed into the original embedding channels (as token features or task-specific relative terms) rather than being modeled as a dedicated positional axis that can be jointly composed with other axes in a unified geometric space, making it less straightforward to scale when multiple axes (i.e., heterogeneous modalities) must be handled together.

A promising trend is multimodal RoPE [29], [30], [31], which treats multiaxis layouts (e.g., text–image or time–spatial) as a unified positional structure and studies how position design and frequency allocation affect cross-axis

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modeling [32]. MHRoPE [29] provides a systematic analysis of multimodal RoPE along these two factors and proposes multihead RoPE to improve full-spectrum utilization and text-prior compatibility. In video contexts, VideoRoPE [30] further analyzes what constitutes a useful video rotary encoding for long-range generalization, and HoPE [31] proposes a hybrid allocation with dynamic temporal scaling for long-context vision–language models. Despite their progress, existing multimodal RoPE designs are still primarily driven by spatiotemporal or vision–language structures and thus do not explicitly model modality identity as a geometric axis, a different type of extra dimension that is crucial for multispectral detection under imperfect alignment. This gap motivates our design of a unified positional encoding that jointly represents spatial location and modality identity in a 3-D positional space.

To the best of our knowledge, no prior work has exploited a 3-D positional embedding space to enhance multimodal feature interaction. To realize this, we propose a unified position embedding strategy that simultaneously encodes both spatial and modality information. Specifically, we introduce MRoPE, which extends 2-D-RoPE by assigning an independent dimension along the modality axis for each modality and defining cross-modal mixed-axis frequencies to instantiate a 3-D rotary embedding. Consequently, the model can explicitly quantify distances between cross-modal tokens, thereby improving the model's awareness of cross-modal position awareness.

B. Attention Mechanism of Transformer

The self-attention mechanism enables the model to capture long-range relationships between tokens through query–key–value (QKV) interactions and has driven substantial advances in deep learning [35], [36]. In the vanilla Transformer [11], QKV are generated via multilayer perceptron projections. Subsequent works have explored alternative QKV generation schemes, such as employing CNNs in place of MLPs, to better adapt the attention mechanism to image processing tasks [37].

In multimodal contexts, CA has emerged as a key approach for coupling heterogeneous features. It can be implemented by taking the query from one modality and the key–value pairs from another, or by jointly forming *QKV* from multiple modalities to enable bidirectional feature exchange. Representative work like MMCA [38], where image queries attend to textual keys to learn soft correspondence matrices. CAFF-DINO [39] couples visible and thermal streams through a hierarchical CA feature-fusion module that injects thermal queries into the visible key–value space at every pyramid level. Inspired by such designs, we embed the proposed MRoPE into the CA computation to form the MRoCA module, enabling the model to capture and quantify intermodal relationships after independent and mixed positional encoding more effectively.

C. Multispectral Object Detection

In multispectral detection, most researchers focus on feature-level fusion techniques to merge features extracted from multimodal inputs to predict bounding boxes [40], [41], [42]. Feature fusion can occur at different network depths, where shallow layers preserve high-resolution features with accurate spatial localization but suffer from limited receptive fields, resulting in weak semantic abstraction and sensitivity to noise. In contrast, Fusion at deeper layers captures richer semantic information but comes at the cost of lower spatial resolution and loss of fine details perception [43], [44].

Representative feature-level fusion methods differ mainly in the fusion position in the model and how they interact and cope with misalignment and modality disparity. C^2 Former [45] focuses on mid-deep layers, employing an intermodal CA module to align and complement RGB-IR features, paired with an Adaptive Feature Sampling front-end that controls computational costs by down-sampling features, effectively addressing modality miscalibration and fusion imprecision. ICAFusion [46] further tightens the tradeoff by introducing dual CA transformers that exchange queries only from the auxiliary modality, followed by an iterative parameter-sharing loop that repeats the same lightweight block; this yields global receptive fields without the typical quadratic expansion of stacked encoders. DAMSDet [47] embeds multispectral deformable CA inside a DETR decoder, cascading queries across three semantic levels and learning sampling offsets that adapt to misalignment at each layer, demonstrating that lateral fusion depth directly modulates the balance between detail preservation and semantic abstraction. Likewise, MiPa [48] revisits the fusion site from a patch-wise perspective, stochastically mixing RGB and IR patches during training and employing a gradient-reversal-based modality-agnostic loss to enforce a single shared encoder that remains invariant to the dominant modality, thereby proving that the fusion granularity itself can serve as regularization rather than architectural overhead. Fusion-Mamba [49] moves the interaction into a latent state space: its fusion-mamba block first swaps channels between modalities for shallow fusion and later performs deep fusion via the dual state space fusion module, compressing modality disparity while preserving spatial fidelity. Overall, these works establish a coherent axis: whether interaction occurs in token, latent-state, or patch space, its resolution, depth, and recurrence jointly govern how well cross-modal features interact and how efficiently complementary cues are distilled.

III. Proposed Method

In this section, we propose a novel Transformer architecture based on adaptive 3-D RoPE, termed AMRo-Former. The overall pipeline is shown in Fig. 2. AMRo-Former incorporates a dual-stream network together with a cross-modal fusion module. For clarity, we partition the cross-modal fusion position along the two-stream backbone into five stages (S_1–S_5). Each stage of fusion comprises two components: MRoCA and AISM.

MRoCA: For each stage, S_k, MRoCA first applies a mixed-axis 3-D rotary embedding (MRoPE) that couples spatial (xy) with the modality axis (z). As a result of introducing the modality axis (z), an intermodal spacing Δz emerges. This Δz quantifies the relative distance between cross-modal tokens and modulates the interaction intensity. After features

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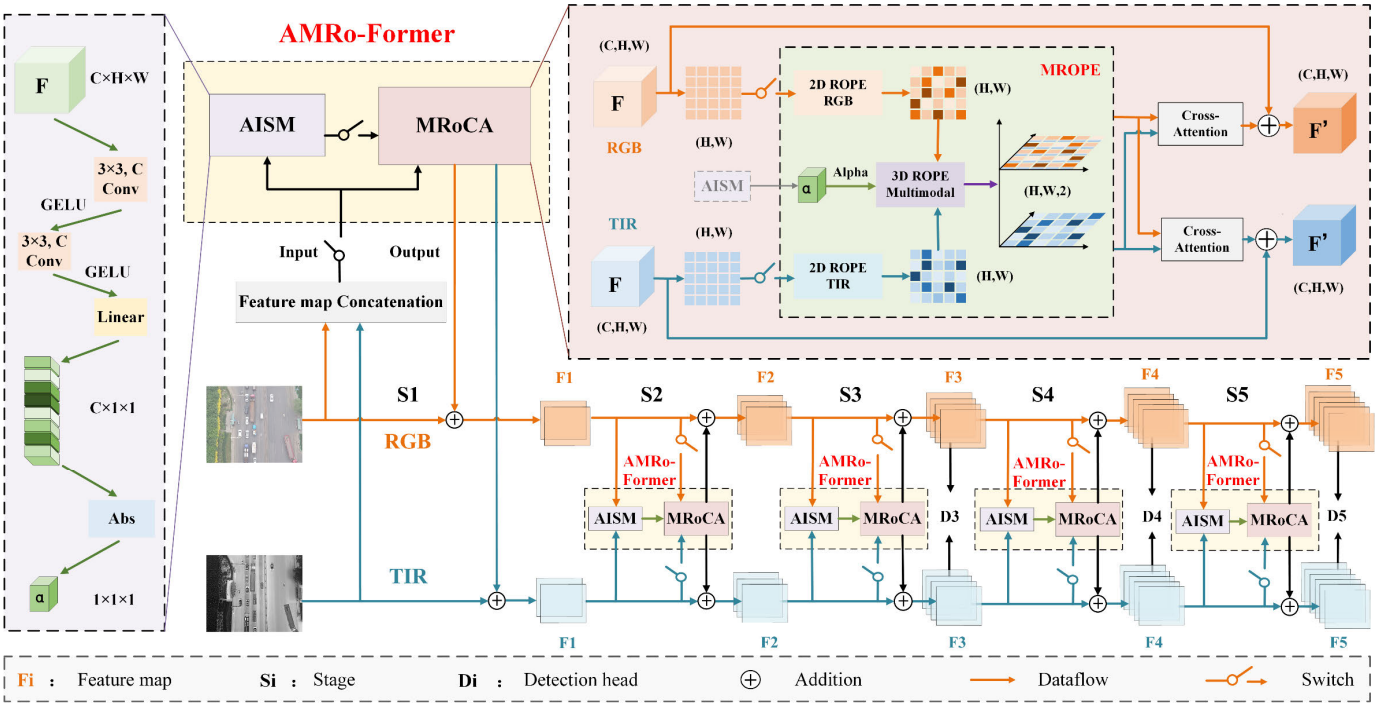


Fig. 2. Backbone framework integrates AMRo-Former, which consists of AISM and MRoCA, with the dual-stream backbone network. AISM processes multimodal features to adaptively derive the optimal solution for inner spacing. The adaptive positional spacing and multimodal features are then input into the MRoCA for multimodal RoPE and CA processing. We deploy switches in front of both AISM and MRoCA to indicate that AISM selection and AMRo-Former selection are possible, enabling further structural exploration for different downstream tasks.

are embedded with both spatial and modality positions, we perform CA: keys/values from one modality attend to queries from the other (optionally symmetrically across branches), enabling explicit, distance-aware interaction between modalities. The fused outputs are passed to the next stage, where the process repeats from S1 to S5.

Adaptive Intermodal Spacing Module: Given stage-wise RGB/IR feature maps, AISM is a lightweight predictor that estimates the optimal intermodal spacing Δz. The predicted Δz is used as auxiliary positional information: we inject it as a third-dimensional (modality axis) input into the subsequent attention module. This design keeps overhead minimal while adapting the strength of cross-modal interaction to the data.

A. 3-D Multimodal RoPE

APE [11], initially used in the Transformer as a fixed function method, employs a combination of sine and cosine functions to encode each position. For each position p and each dimension d, the encoding can be expressed as follows:

$$APE\,(p, 2i) = \sin\left(\frac{p}{10000^{(i/d)}}\right) \tag{1}$$

$$APE\,(p, 2i+1) = \cos\left(\frac{p}{10000^{(i/d)}}\right) \tag{2}$$

where APE represents the positional encoding matrix, p is the position index, i is the dimension index, and d is the total number of encoding dimensions. APE encodes positions in absolute coordinates. Consequently, it extrapolates poorly to out-of-range positions and lacks distance awareness between tokens.

To address this limitation, researchers have introduced RPE [33], which mitigates the constraints of absolute positional encoding in handling long sequences. Traditional methods of RPE compute the relative position between token m and n using the following formula:

$$RPE\,(p_m, p_n) = \sin\left(\frac{p_m - p_n}{10000^{(i/d)}}\right) \tag{3}$$

where p_m and pn are the indices of two positions. While RPE enables more efficient utilization of relative positional information, it involves more complex processing and higher computational costs, thus requiring compatible mechanisms to implement the encoding effectively.

RoPE [18] embeds the position information of input I_m and I_n by rotating each 2-D subvector of queries/keys with a phase that is linear in position

$$f_q\,(I_m, p_m) = \left(W_m^q I_m\right) e^{ip_m\theta} = \left(W_m^q I_m\right) R_m \tag{4}$$

$$f_k\,(I_n, p_n) = \left(W_n^k I_n\right) e^{ip_n\theta} = \left(W_n^k I_n\right) R_n \tag{5}$$

where W^q and W^k represent projection matrices for queries and keys, respectively, and θ denotes multiple frequencies. The $e^{ip_m\theta}$ and $e^{ip_n\theta}$ are denoted as a rotation matrix R. The attention matrix $\mathcal{A}$ between query and key is computed as

$$\mathcal{A}\,(Q_m, K_n) = \mathrm{Re}\left[\left(W_m^q I_m\right)^{\mathrm{T}} \left(W_n^k I_n\right)^* e^{i(p_m - p_n)\theta}\right] \tag{6}$$

where Re[·] extracts the real part of a complex number, and $(\cdot)^*$ denotes the complex conjugate. Therefore, we obtain

$$\mathcal{A}\,(Q_m, K_n) \propto_{p_m - p_n} \mathrm{Re}\left(e^{i(p_m - p_n)\theta}\right). \tag{7}$$

Therefore, from (7), we can see that the attention matrix is directly related to the relative positions of tokens $(p_m - p_n)$.

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RoPE was originally proposed for 1-D text, and recent works have validated 2-D instantiation for vision tasks. To address this, 2-D RoPE [20] assigns 2-D position indices, therefore encoding spatial relationships in image data. However, 2-D RoPE is fundamentally unimodal, because it encodes only spatial positions and cannot express the relationship between modalities, leaving the attention mechanism unaware of the cross-modal information.

A straightforward baseline to inject modality information is to add a learnable modality type embedding vector (τ_{rgb} and τ_{ir}) to each token. Concretely, let x_i^m be a token from modality $m \in \{RGB, IR\}$. With type embeddings, the projected queries/keys become $q_i = (x_i^m + \tau_m)W_q$ and $k_j = (x_j^n + \tau_n)W_k$, and the attention logit expands as

$$q_i^T k_j = \left(x_i^m W_q\right)^T \left(x_j^n W_k\right) + \left(x_i^m W_q\right)^T (\tau_n W_k) + \left(\tau_m W_q\right)^T \left(x_j^n W_k\right) + \left(\tau_m W_q\right)^T (\tau_n W_k) \tag{8}$$

where the only purely modality-pair dependent component is $(\tau_m W_q)^T(\tau_n W_k)$, which behaves as a constant bias for each modality pair and does not represent a continuous "modality distance" that can be jointly composed with spatial offsets. In contrast, RoPE-style encodings inject relative structure directly into attention via rotations. When we allocate an explicit modality coordinate, p^z, and apply az-axis rotation $e^{ip^z\theta}$, the cross-modal interactions are modulated by a relative rotation, which satisfies

$$\mathcal{A}(Q_m, K_n) \propto_{p_m^z - p_n^z} \mathrm{Re}\left(e^{i(p_m^z - p_n^z)\theta}\right) \tag{9}$$

where p^z denotes the modality coordinate of each modality. Equation (7) provides a distance-aware and translation-equivariant attention matrix to encode modality relationships that cannot be reproduced by a single additive-type vector.

To address this, we design a 3-D positional encoding strategy. Our motivation is to make cross-modal distances quantifiable at the encoding stage. We augment the spatial position plane xy with an orthogonal modality axis z and assign each modality a coordinate p^z. Thus, our 3-D position indices of any two tokens are $p_m = (p_m^x, p_m^y, p_m^z)$, $p_n = (p_n^x, p_n^y, p_n^z)$, where $p^x \in \{1, 2, \ldots, w\}$ and $p^y \in \{1, 2, \ldots, h\}$ are spatial indices. For the modality axis, an initial assignment $p^z \in (1, 2)$ reuses the unit step of the (x, y) grid and implicitly fixes the intermodal spacing Δz to $\Delta z = |p_m^z - p_n^z| = 1$. However, the definition of spatial grid and intermodal spacing is inherently distinct; enforcing such a setting is overly naive and ignores their fundamental differences. To decouple modality spacing from spatial step, we introduce a spacing factor α and redefine the index

$$\tilde{p}^z = \alpha \times p^z, p^z \in (1, 2). \tag{10}$$

Thus, the intermodal spacing between cross-modal tokens is $\Delta z = |\tilde{p}_m^z - \tilde{p}_n^z| = \alpha$. Intuitively, the scaling coefficient α governs the relative distances between cross-modal tokens, thereby influencing the intensity of cross-modal interactions. Through the following experiments, we observe that the variations in α induce significant changes in model performance. Building upon this, we introduce our mixed-axis 3-D rotary embedding (MRoPE), where the rotation matrix R is defined as a mixed-frequency rotation matrix

$$R(m) = e^{i\left(\theta^x p_m^x + \theta^y p_m^y + \theta^z \alpha p_m^z\right)} \tag{11}$$

where θ^x, θ^y, and θ^z are base angular frequencies for the x-, y-, and z-axes, respectively. In standard RoPE, each attention head of width d_{head} is partitioned into $d_{head}/2$ 2-D subsectors indexed by $t \in (0, 1, \ldots, d_{head}/2 - 1)$ to achieve a rotatable plane. In our MRoPE, we group these subvectors into triplets: one plane per axis indexed by t, where $t \in (0, 1, \ldots, d_{head}/6 - 1)$, yielding indices $3t$, 3t + 1, 3t + 2 for x, y,and z planes, respectively. Therefore, the rotary matrix with these indices is

$$\begin{aligned} R(m, 3t) &= e^{i\theta_t p_m^x}, R(m, 3t+1) \\ &= e^{i\theta_t p_m^y}, R(m, 3t+2) = e^{i\theta_t \alpha p_m^z}. \end{aligned} \tag{12}$$

According to (6), it can be easily derived that

$$\mathcal{A}(I_m, I_n) \propto_{p_m^x - p_n^x} \mathrm{Re}\left(e^{i(p_m^x - p_n^x)\theta_t^x}\right) \tag{13}$$

$$\mathcal{A}(I_m, I_n) \propto_{p_m^y - p_n^y} \mathrm{Re}\left(e^{i(p_m^y - p_n^y)\theta_t^y}\right) \tag{14}$$

$$\mathcal{A}(I_m, I_n) \propto_{\alpha(p_m^z - p_n^z)} \mathrm{Re}\left(e^{i\alpha(p_m^z - p_n^z)\theta_t^z}\right). \tag{15}$$

This approach allows MRoPE to capture relative positional information between different modalities while maintaining xy spatial position information. The specific structure is illustrated in Fig. 3.

B. Adaptive Intermodal Spacing Module

In MRoPE, the modality axis contributes an intermodal spacing term Δz, which directly controls how strongly features from different modalities interact in CA and therefore must be optimized. To eliminate the need for grid search while preserving efficiency, we introduce a lightweight AISM that automatically infers the optimal Δz. As illustrated in Fig. 2, AISM is a lightweight 2-D CNN module. Its core structure efficiently maps complex feature inputs to a 1-D scalar output through hierarchical spatial compression and feature distillation. Specifically, AISM first splits the feature map output by the backbone network along the batch dimension into RGB modality features and IR modality features. These two different modality features are concatenated to form a hybrid feature containing multimodal information. AISM then applies two sequential 3×3 convolutional layers to extract features from the hybrid feature, with each convolutional layer followed by a GELU activation function to enhance the model's nonlinear expression capability:

$$\mathbf{Y}_1 = \mathrm{GELU}\left(\mathrm{Conv2D}_{3\times3}^{3,1}(\mathbf{X}; \mathbf{W}_1)\right) \tag{16}$$

$$\mathbf{Y}_2 = \mathrm{GELU}\left(\mathrm{Conv2D}_{3\times3}^{3,1}(\mathbf{Y}_1; \mathbf{W}_2)\right) \tag{17}$$

where $\mathbf{X}$ represents the input hybrid features, GELU $(\cdot)$ denotes the Gaussian Error Linear Unit activation function, and $\mathrm{Conv2D}_{3\times3}^{3,1}$ is the 2-D convolution operator. After applying the convolution, the features are compressed into a 1-D vector. This vector is then passed through a fully connected layer to map it to a scalar output, which is subsequently summed

$$h = \mathrm{Flatten}(\mathbf{Y}_2)\mathbf{W}_3 + \beta_3 \tag{18}$$

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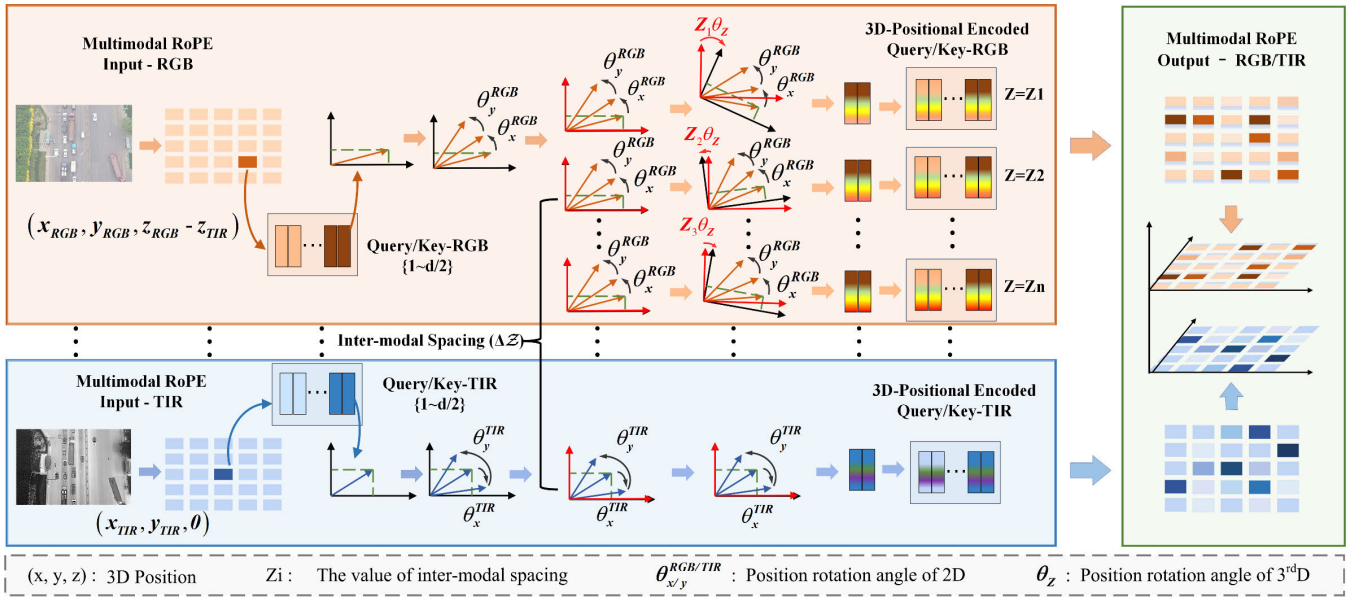


Fig. 3. 3-D RoPE architecture applied to MRoPE. We set the infrared modality position to 0; at this point, the visible modality position corresponds to the intermodal spacing. Due to the variability in modal positions (Z), different values of intermodal spacing (ΔZ) will generate varying degrees of modal position rotary embedding. The red angle represents the modal rotary.

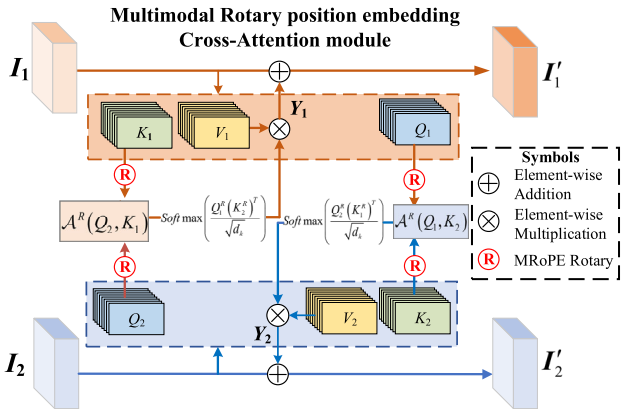


Fig. 4. Multimodal rotary position embedding cross-attention (MRoCA).

where Flatten (·) represents the flattening operation. After flattening, the features are passed through the GELU activation function, which aggregates the feature information to achieve the optimal intermodal spacing solution. The aggregation process allows the model to efficiently learn the best setting for intermodal spacing by leveraging the batch's collective feature representations

$$\alpha = \mathrm{GELU}\left(\sum_{i=1}^{H} |h_i|\right). \tag{19}$$

C. Adaptive Multimodal RoPE Transformer

Both AISM and MRoPE aim to supply explicit cross-modality positional information: AISM estimates an optimal intermodal spacing factor α, while MRoPE embeds both spatial and modal position information. Since our final goal is feature fusion, we embed MRoPE into a CA operator, yielding the MRoCA module (as shown in Fig. 4). We then place AISM and the MRoCA module inside a dual-stream backbone stage by stage to develop an AMRo-Former, to validate its effectiveness in the multispectral object detection task.

Specifically, for branch 1 and branch 2 inputs, I_1 and I_2 (from the RGB and TIR streams), which are transformed into queries, keys, and values

$$Q_j = I_j W^{Q_j},\ K_j = I_j W^{K_j},\ V_j = I_j W^{V_j} \tag{20}$$

where $j \in (1,2)$ denotes the branch index, each corresponding to a distinct p^z. The optimal intermodal data solution α generated by AISM is incorporated into the rotary embedding function R. The QKV is processed using R to perform MRoPE

$$Q_j^R = Q_j R(m,t) = Q_j e^{i\left(\theta_t^x p_m^x + \theta_t^y p_m^y + \theta_t^z \alpha j\right)} \tag{21}$$

$$K_j^R = K_j R(n,t) = K_j e^{i\left(\theta_t^x p_n^x + \theta_t^y p_n^y + \theta_t^z \alpha j\right)}. \tag{22}$$

After obtaining multimodal positional information Q_i^R and K_i^R, we employ a multihead CA mechanism, as shown in Fig. 4. Let d be the embedding dimension per token, d_h the per-head width. We define the two bidirectional CAs (branch $1 \rightarrow 2$ and $2 \rightarrow 1$) as

$$\begin{aligned}\mathcal{A}^R &= \left\{\mathcal{A}_1^R\left(Q_2^R, K_1^R\right), \mathcal{A}_2^R\left(Q_1^R, K_2^R\right)\right\} \\ &= \left\{\mathrm{softmax}\left(\frac{Q_2^R\left(K_1^R\right)^T}{\sqrt{d_k}}\right), \mathrm{softmax}\left(\frac{Q_1^R\left(K_2^R\right)^T}{\sqrt{d_k}}\right) V_{RGB}\right\}.\end{aligned} \tag{23}$$

Given the attention weights, the multihead responses are

$$\begin{aligned}Y_1 &= \left[\mathrm{Concat}_{h=1}^{H}\left(\mathcal{A}_1^{R,(h)} V_1\right)\right] W_1, Y_2 \\ &= \left[\mathrm{Concat}_{h=1}^{H}\left(\mathcal{A}_2^{R,(h)} V_2\right)\right] W_2\end{aligned} \tag{24}$$

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where W_1 and W_2 denote the output projection matrix for the multihead response. Finally, we use a residual connection to add the multihead response back into the input of each branch

$$I_1' = I_1 + Y_1, I_2' = I_2 + Y_2. \tag{25}$$

If traditional CA is used, then according to (23), we have

$$\begin{aligned}\mathcal{A} &= \{\mathcal{A}_1(Q_2, K_1), \mathcal{A}_2(Q_1, K_2)\} \\ &= \left\{\text{softmax}\left(\frac{Q_2 (K_1)^T}{\sqrt{d_k}}\right), \text{softmax}\left(\frac{Q_1 (K_2)^T}{\sqrt{d_k}}\right) V_{RGB}\right\}\end{aligned} \tag{26}$$

$$\begin{aligned}Y_1^{\text{CA}} &= \left[\text{Concat}_{h=1}^{H}\left(\mathcal{A}_1^{(h)} V_1\right)\right] W_1, Y_2^{\text{CA}} \\ &= \left[\text{Concat}_{h=1}^{H}\left(\mathcal{A}_2^{(h)} V_2\right)\right] W_2,\end{aligned} \tag{27}$$

$$I_1' = I_1 + Y_1^{\text{CA}}, I_2' = I_2 + Y_2^{\text{CA}}. \tag{28}$$

If the MRoCA or any other CA module is disabled, the branch feature is forwarded unchanged with no residual update

$$I_1' = I_1, I_2' = I_2. \tag{29}$$

IV. Experimental Results

A. Experimental Dataset Description and Training Details

Since our algorithm is designed for object detection in dual-spectrum environments, it is evaluated on the VEDAI [50], drone-vehicle [42], and LLVIP [51] datasets.

1) *Vehicle Detection in Aerial Imagery:* VEDAI is a remote sensing dataset for VEDAI, comprising 3640 vehicle instances across nine categories, including ships, cars, RVs, airplanes, shuttle buses, tractors, trucks, vans, and others. The dataset includes 1210 aerial images from AGRC in Utah, with a spatial resolution of 12.5 cm. VEDAI has usually been used for benchmarking automatic target recognition algorithms in unconstrained environments.
2) *Drone-Vehicle:* The drone-vehicle dataset comprises 56 878 images collected by drones, with 28 439 pairs of RGB and infrared images, encompassing five categories. Specifically, the remote sensing dataset includes 389 779 RGB annotations and 428 086 infrared annotations for cars, 22 123 RGB annotations and 25 960 infrared annotations for trucks, 15 333 RGB annotations and 16 590 infrared annotations for buses, 11 935 RGB annotations and 12 708 infrared annotations for vans, and 13 400 RGB annotations and 17 173 infrared annotations for freight cars.
3) *Low-Light Vision Infrared-Visible Paired:* LLVIP dataset is a visible-infrared paired dataset for low-light vision, containing 30 976 images, i.e., 15 488 pairs, most of which are captured in scenes with extremely low visibility, with all images strictly aligned in both time and space.

B. Training Details and Evaluation Metrics

We conducted all experiments with the stochastic gradient descent (SGD) optimizer, an initial learning rate of $1e^{-2}$, momentum of 0.937, and weight decay of 0.0005. Our algorithm was implemented on an 80 GB NVIDIA[1] Tesla A100 Tensor Core GPU, with training performed for 200 epochs on the input size of 640 × 640. We utilized a YOLOv5L detector and a pretrained model on the COCO dataset for initializing the weights. We use mAP as the primary metric, including mAP50, mAP75, and mAP (average over IoU 0.5:0.95). Precision (positive predictive value) and Recall are also reported. We additionally compare model complexity via the number of parameters [Param (m)] and GFLOPs. We further evaluated the frames per second (FPS) of all algorithms on the NVIDIA Jetson AGX Orin platform, without employing any acceleration techniques. FPS is averaged over ten runs after warm-up and includes postprocessing such as NMS.

C. Comparisons With Previous Methods

We present the experimental results of our algorithm for dual-spectrum (visible and infrared) object detection on three datasets: VEDAI, drone-vehicle, and LLVIP. We compare our designed model with the SOTA methods in single-modal algorithms (including YOLOv8, YOLOv9 [52], GHOST [53]) and multimodal algorithms (such as Gliding Vertex [54], C2Former [45], MiPA [48], SUPER-YOLO [55], ICAFusion [46], DAMSDet [47], LF-MDet [56], DUAL-YOLOv9, DAC-Fusion [57], CFT [58], and OAFA [59]). To balance the algorithm's performance and parameter count, our general model in experiments adopts the following configuration: the MRoCA module [see (25)] is used in S1, S3, and S4; the standard CA module [see (28)] is used in S5; and no module is employed in S2 [see (29)]. In addition, we also test a variant without AISM for comparative purposes, with the α parameter in MRoCA (for S1, S3, and S4) fixed at 1.5.

Fig. 5 demonstrates the detection performance of our algorithm on the respective datasets, while Tables I–IV present the detailed detection metrics of our method and other comparative methods. Here, "(α = 1.5)" denotes that α is set to 1.5, while "(AISM)" denotes that α is adaptively determined by the AISM module. All datasets are split into training and validation sets at an 8:2 ratio. The comparison results are summarized as follows.

1) *VEDAI Dataset:* Due to the smaller scale and diverse scenes of the VEDAI dataset compared to typical dual-band object detection datasets, training is more challenging. Tables I–IV show that, compared to the optimal SOTA results, our algorithm improves mAP50 by 0.0169 over the DACFusion algorithm and mAP75 by 0.0639 over the CFT algorithm. In comparison to DUAL-YOLOv9C, our model improves mAP and precision by 0.0101 and 0.0447, and recall by 0.0386. Replacing the fixed intermodal spacing with AISM further improves our method, yielding 0.0073, 0.0307, 0.0118 gains in mAP50, mAP75, and mAP compared with the fixed-spacing variant (α = 1.5) (see Table I). In terms of Precision and Recall, AISM increases recall by 0.0122 while slightly reducing precision by −0.0432

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TABLE I

COMPARISON WITH SOTA METHOD FOR THE VEDAI, DRONE-VEHICLE, AND LLVIP DATASETS FOR MAP50, MAP75, AND MAP. "AMRO-FORMER (α = 1.5)" DENOTES THAT INTERMODAL SPACING IS SET TO 1.5. "AMRO-FORMER (AISM)" DENOTES THE INTERMODAL SPACING IS ADAPTIVELY DETERMINED BY AISM

Algorithm			VEDAI			Drone-Vehicle			LLVIP		
	Method	Modal	mAP50	mAP75	mAP	mAP50	mAP75	mAP	mAP50	mAP75	mAP
Single modal	YOLOv8L	IR	0.6798	0.5304	0.4402	0.8038	0.6587	0.5585	0.9758	0.7921	0.6863
	YOLOv8L	RGB	0.7342	0.5581	0.4794	0.8035	0.6776	0.5758	0.9614	0.6836	0.6159
	YOLOv9C (ECCV'24)	IR	0.7424	0.5693	0.4807	0.8149	0.6639	0.5681	0.9858	0.8317	0.7182
	YOLOv9C (ECCV'24)	RGB	0.7841	0.6330	0.5262	0.8214	0.7160	0.6082	0.9644	0.6983	0.6239
	GHOST (TGRS'23)	IR	0.6828	0.5219	0.4339	0.7357	0.5414	0.4763	0.9770	0.7418	0.6482
	GHOST (TGRS'23)	RGB	0.7196	0.5371	0.4650	0.7520	0.5956	0.5085	0.9391	0.6549	0.5411
Multi modal	Gliding Vertex (TPAMI'20)	RGB+TIR	0.7359	0.4594	0.4225	0.6024	0.3886	0.3671	0.8357	0.5112	0.4969
	C^2Former (TGRS'24)	RGB+TIR	0.6770	0.2602	0.3316	0.6918	0.4986	0.4368	0.9310	0.7301	0.5518
	SUPER-YOLO (TGRS'23)	RGB+TIR	0.7443	0.5188	0.4547	0.7237	0.5484	0.4690	0.9688	0.7432	0.6513
	ICAFusion (PR'24)	RGB+TIR	0.7532	0.4443	0.4330	0.8433	0.7605	0.6276	0.9815	0.7855	0.6760
	LF-MDet (TGRS'24)	RGB+TIR	0.8080	-	0.5110	0.7180	-	0.5130	-	-	-
	DUAL-YOLOv9C	RGB+TIR	0.8648	0.6447	0.5845	0.7974	0.6498	0.5992	0.9839	0.7713	0.7114
	DACFusion (IJON'25)	RGB+TIR	0.8720	-	0.5760	-	-	-	0.9790	-	0.6910
	CFT (PR'22)	RGB+TIR	0.8525	0.6676	0.5492	0.8487	0.7461	0.6267	0.9768	0.7782	0.6739
	DAMSDet (ECCV'24)	RGB+TIR	0.7984	0.6006	0.5034	0.817	0.634	0.560	0.9814	0.7832	0.6905
	MiPA (WACV'25)	RGB+TIR	0.7514	0.5063	0.4617	0.8049	0.6235	0.5711	0.9820	0.7810	0.6650
	OAFA (CVPR'24)	RGB+TIR	0.8046	0.6077	0.5119	0.7914	0.6209	0.5583	0.9731	0.7698	0.6691
	AMRo-Former (α = 1.5)	RGB+TIR	0.8889	0.7315	0.5946	0.8520	0.7608	**0.6382**	0.9878	0.7932	0.6942
	AMRo-Former (AISM)	RGB+TIR	**0.8962**	**0.7622**	**0.6064**	**0.8531**	**0.7730**	0.6362	**0.9937**	**0.8984**	**0.7641**

(see Table II), indicating a recall–precision tradeoff on VEDAI.

2) *Drone-Vehicle Dataset:* Tables I–IV demonstrate that, compared to the SOTA results, our algorithm improves mAP50 by 0.0033 over the CFT algorithm, and mAP75, mAP by 0.0003 and 0.0106, over the ICAFusion algorithm. Precision and Recall are improved by 0.0076 and 0.0047 over the C2Former algorithm. These results indicate that our algorithm retains a significant advantage even on this large-scale dataset. Introducing AISM mainly improves mAP75 by 0.0122 and slightly increases mAP50 by 0.0011, while keeping the overall mAP comparable (−0.0020) relative to the fixed-spacing variant (α = 1.5) (see Table I). In terms of precision and recall, AISM yields a small recall gain (+0.0079) with a modest precision decrease (−0.0201) (see Table II).
3) *LLVIP Dataset:* Tables I–IV show that, compared to the optimal SOTA results, our algorithm improves mAP50 by 0.0039 over the DUAL-YOLOv9C algorithm and mAP75 by 0.0077 over the ICAFusion algorithm. However, compared to DUAL-YOLOv9C, mAP decreases by 0.0172. Precision is improved by 0.0033 over the ICAFusion algorithm and Recall by 0.0037 over the DUAL-YOLOv9C algorithm. With AISM, we further improve the fixed-spacing variant (α = 1.5) by 0.0059, 0.1052, and 0.0699 in mAP50, mAP75, and mAP, together with +0.0196 precision and +0.0171 recall (see Tables I and II).

As shown in Fig. 6, we plotted the detection score curves (mAP50 and mAP) over training epochs on the VEDAI, drone-vehicle, and LLVIP datasets using our algorithm and the SOTA methods for comparison. To highlight the algorithm's performance on remote sensing images, the detection score trends for the VEDAI and drone-vehicle datasets are specifically shown. Fig. 7 compares detection performance across the three datasets and includes feature map visualizations for VEDAI and drone-vehicle. Fig. 8 further illustrates the performance differences across the datasets for our method and each SOTA method.

Moreover, the proposed AMRo-Former achieves 15 FPS, which is comparable to transformer-based fusion baselines (e.g., CFT) reported in Tables III and IV. This runtime indicates that, on modern edge GPU platforms typically used for UAV payloads, the computational profile of AMRo-Former is practically manageable for on-board perception. Meanwhile, the current implementation is not yet optimized for deployment (e.g., TensorRT compilation, FP16/INT8 quantization), and

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TABLE II

COMPARISON WITH SOTA METHOD FOR THE VEDAI, DRONE-VEHICLE, AND LLVIP DATASETS FOR PRECISION AND RECALL. "AMRO-FORMER (α = 1.5)" DENOTES THAT INTERMODAL SPACING IS SET TO 1.5. "AMRO-FORMER (AISM)" DENOTES THE INTERMODAL SPACING IS ADAPTIVELY DETERMINED BY AISM

Algorithm			VEDAI		Drone-Vehicle		LLVIP	
	Method	Modal	Precision	Recall	Precision	Recall	Precision	Recall
Single modal	YOLOv8L	IR	0.7518	0.6089	0.7703	0.7858	0.9568	0.9630
	YOLOv8L	RGB	0.7158	0.7173	0.7694	0.7729	0.9373	0.9222
	YOLOv9C (ECCV'24)	IR	0.7311	0.7132	0.7805	0.7804	0.9507	0.9589
	YOLOv9C (ECCV'24)	RGB	0.7394	0.7490	0.8044	0.7759	0.9409	0.9264
	GHOST(TGRS'23)	IR	0.6873	0.6443	0.7184	0.7357	0.9480	0.9459
	GHOST(TGRS'23)	RGB	0.6920	0.6326	0.7336	0.7310	0.9239	0.8999
Multi modal	Gliding Vertex (TPAMI'20)	RGB+TIR	0.7520	0.7212	0.8080	0.7789	0.8696	0.8845
	C^2Former (TGRS'24)	RGB+TIR	0.7110	0.6585	0.8481	0.8195	0.9370	0.9162
	SUPER-YOLO (TGRS'23)	RGB+TIR	0.6750	0.7381	0.7038	0.7008	0.9160	0.9583
	ICAFusion (PR'24)	RGB+TIR	0.7982	0.6979	0.8183	0.7976	0.9603	0.9562
	DUAL-YOLOv9C	RGB+TIR	0.8385	0.7965	0.7578	0.7644	0.9463	0.9604
	CFT (PR'22)	RGB+TIR	0.8187	0.7985	0.8300	0.7965	0.9480	0.9426
	DAMSDet (ECCV'24)	RGB+TIR	0.7206	0.6254	0.8094	0.7234	0.9061	0.9345
	MiPA (WACV'25)	RGB+TIR	0.7659	0.7085	0.8087	0.7823	0.9617	0.9547
	OAFA (CVPR'24)	RGB+TIR	0.8128	0.7396	0.7879	0.7531	0.9530	0.9445
	AMRo-Former (α = 1.5)	RGB+TIR	**0.8832**	0.8351	**0.8557**	0.8242	0.9636	0.9641
	AMRo-Former (AISM)	RGB+TIR	0.8400	**0.8473**	0.8356	**0.8321**	**0.9832**	**0.9812**

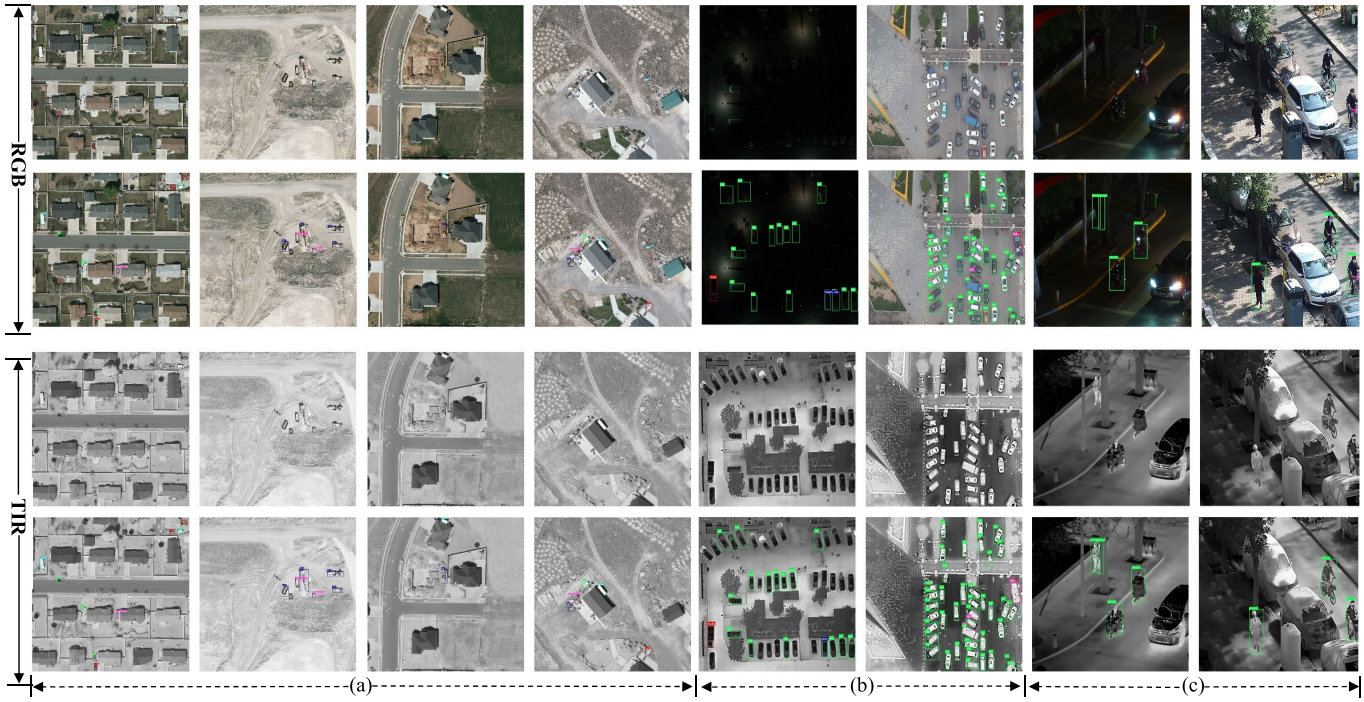


Fig. 5. Dual-spectrum object detection results for typical categories across VEDAI, drone-vehicle, and LLVIP datasets. Rows: top two rows show RGB images (first row: original; second row: with detection boxes); and bottom two rows show TIR images (third row: original; fourth row: with detection boxes). (a) VEDAI dataset, (b) drone-vehicle dataset, and (c) LLVIP dataset.

further speed-ups are expected with standard engineering practice. In addition, we have conducted preliminary architectural instantiations in the ablation/efficiency analysis (Section IV-D), showing that convolution-augmented

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TABLE III
PER-CLASS DETECTION PERFORMANCE ON THE VEDAI DATASET

Method	Modal	car	boat	pickup	van	tractor	camp	other	truck	mAP50	Param(m)	GFLOPs	FPS
YOLOv8L	IR	0.890	0.680	0.826	0.606	0.600	0.759	0.713	0.366	0.680	43.6	164.8	17
YOLOv8L	RGB	0.888	0.751	0.802	0.687	0.819	0.752	0.639	0.532	0.734	43.6	164.8	17
YOLOv9C	IR	0.912	0.690	0.850	0.722	0.713	0.850	0.737	0.466	0.742	50.7	236.7	16
YOLOv9C	RGB	0.921	0.786	0.842	0.679	0.854	0.889	0.702	0.597	0.784	50.7	236.7	16
GHOST	IR	0.888	0.334	0.861	0.644	0.634	0.840	0.642	0.620	0.683	-	-	20
GHOST	RGB	0.940	0.439	0.876	0.625	0.909	0.862	0.509	0.596	0.720	-	-	20
Gliding Vertex	RGB+TIR	0.872	0.721	0.835	0.682	0.817	0.820	0.562	0.578	0.736	41.1	107.2	10
C^2Former	RGB+TIR	0.842	0.698	0.773	0.610	0.759	0.740	0.486	0.506	0.677	101.0	258.3	21
SUPER-YOLO	RGB+TIR	0.875	0.712	0.850	0.693	0.789	0.773	0.686	0.567	0.744	7.0	81.5	18
ICAFusion	RGB+TIR	0.898	0.720	0.807	0.688	0.835	0.814	0.679	0.585	0.753	120.2	180.0	17
LF-MDet	RGB+TIR	0.921	0.688	0.873	0.841	0.895	0.859	0.671	0.716	0.808	38.7	77.7	-
DUAL-YOLOv9C	RGB+TIR	0.944	0.733	0.934	0.855	0.936	0.922	0.751	0.845	0.865	50.7	237.4	16
CFT	RGB+TIR	0.939	0.784	0.876	0.837	0.939	0.927	0.694	0.840	0.853	206.0	224.4	15
DAMSDet	RGB+TIR	0.937	**0.829**	0.830	0.883	0.642	0.667	0.728	0.870	0.798	79.07	83.0	-
MiPA	RGB+TIR	0.892	0.686	0.846	0.679	0.838	0.815	0.664	0.585	0.751	217.8	139.6	-
OAFA	RGB+TIR	0.91	0.722	0.862	0.812	0.905	0.869	0.651	0.706	0.805	-	-	-
AMRo-Former ($\alpha = 1.5$)	RGB+TIR	0.969	0.721	0.949	**0.951**	0.960	0.937	**0.755**	**0.873**	0.889	264.8	224.8	15
AMRo-Former (AISM)	RGB+TIR	**0.975**	0.798	**0.959**	0.942	**0.965**	**0.941**	0.700	0.830	**0.896**	265.1	226.5	15

lightweight variants can substantially reduce Params/GFLOPs with a moderate accuracy tradeoff, which we believe is a valuable and recommended direction for stricter real-time constraints.

In conclusion, the exceptional performance achieved on the VEDAI and drone-vehicle datasets demonstrates that our design exhibits significant multimodal processing capabilities in the domain of remote sensing. Furthermore, the superior performance suggests that our design is not limited to specific application domains, and we anticipate a wider range of applications of multimodal tasks.

D. Experimental Ablation Result

To validate the value and necessity of MRoCA and AISM in multispectral object detection, we conducted an ablation study on the VEDAI dataset, training for 150 epochs with an input size of 640 × 640. Unless otherwise specified, all ablation baselines use the same dual-stream backbone and the same detection head/neck. The only difference lies in the fusion block and positional encoding (The specific definitions of the baseline are provided in Table V).

1) *Effectiveness of MRoCA and AISM:* We tested the detection performance when MRoCA and AISM are inserted into all stages (S1–S5) of AMRo-Former and compared the results against ablative settings at those stages: using only MRoCA (with fixed $\alpha = 1.5$), using a standard CA module [see (28)]. The results are shown in Table VI. We found that equipping the model with MRoCA + AISM achieves the best results in mAP50, mAP75, and mAP. The standard CA module achieved the lowest scores in mAP50, mAP75, and mAP. We argue that this does not indicate that the CA module has a negative effect, but rather that traditional CA may exhibit low training efficiency on image data. These results validate that integrating the MRoCA and AISM modules can significantly enhance multimodal object detection performance. The experimental results are also visualized in Fig. 9.
2) *Performance Assessment of Positional Encoding Method:* We have developed the MRoPE, which extends the RoPE method to encode both modal and spatial information of features. To investigate the impact of MRoPE on the model's performance within MRoCA, we conducted ablative experiments with four settings: MRoCA using our proposed MRoPE, MRoCA using 2-D RoPE, the standard CA module with traditional APE, and a hybrid strategy where 2-D RoPE is used for spatial encoding while a learnable APE is adopted for modality encoding (denoted as 2-D RoPE + APE). Note that there are only two modality indices under the RGB–IR setting, and therefore, using APE along the modality axis reduces to a standard modality type embedding baseline. Table VII directly compares z-axis rotary modeling (MRoPE) against

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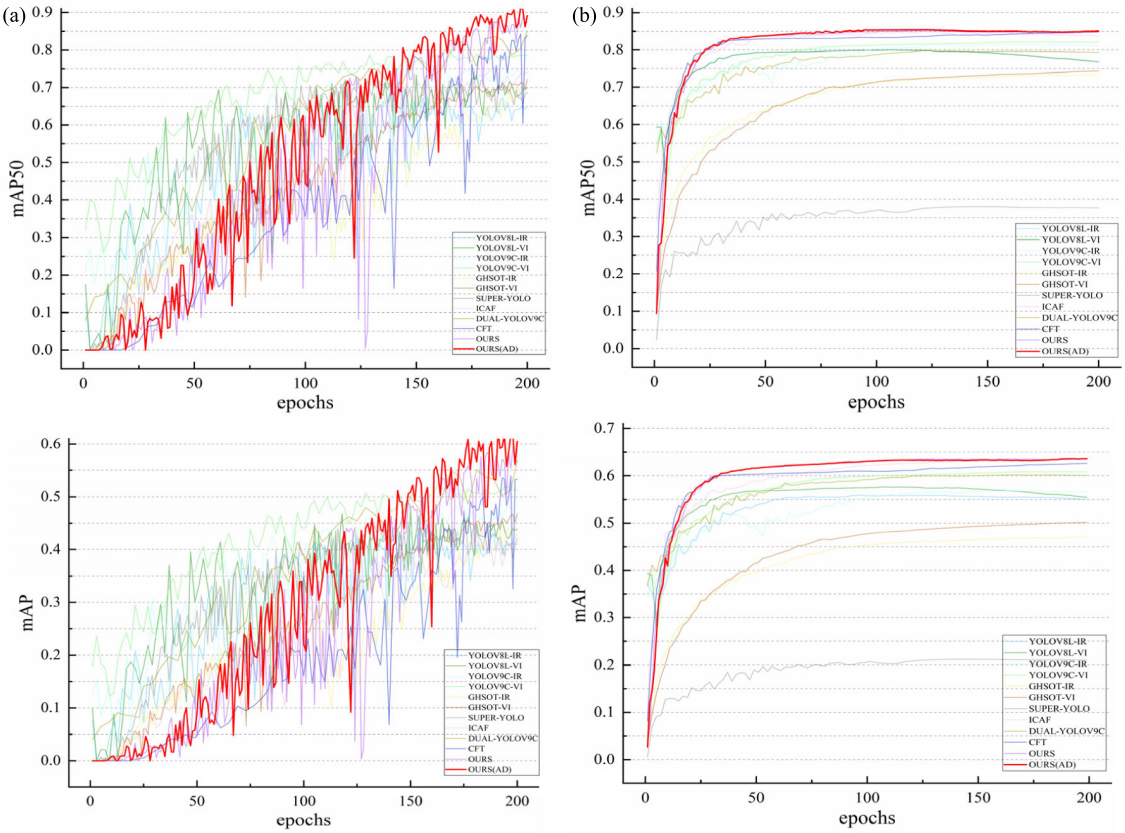


Fig. 6. Comparison of target detection scores between our algorithm and SOTA on the VEDAI and drone-vehicle remote sensing datasets. Specifically, (a) VEDAI dataset, while (b) drone-vehicle dataset. The upper plots show the mAP50 score curves, and the lower plots display the mAP score curves. The red curve corresponds to our algorithm, while the other colors represent the compared algorithms.

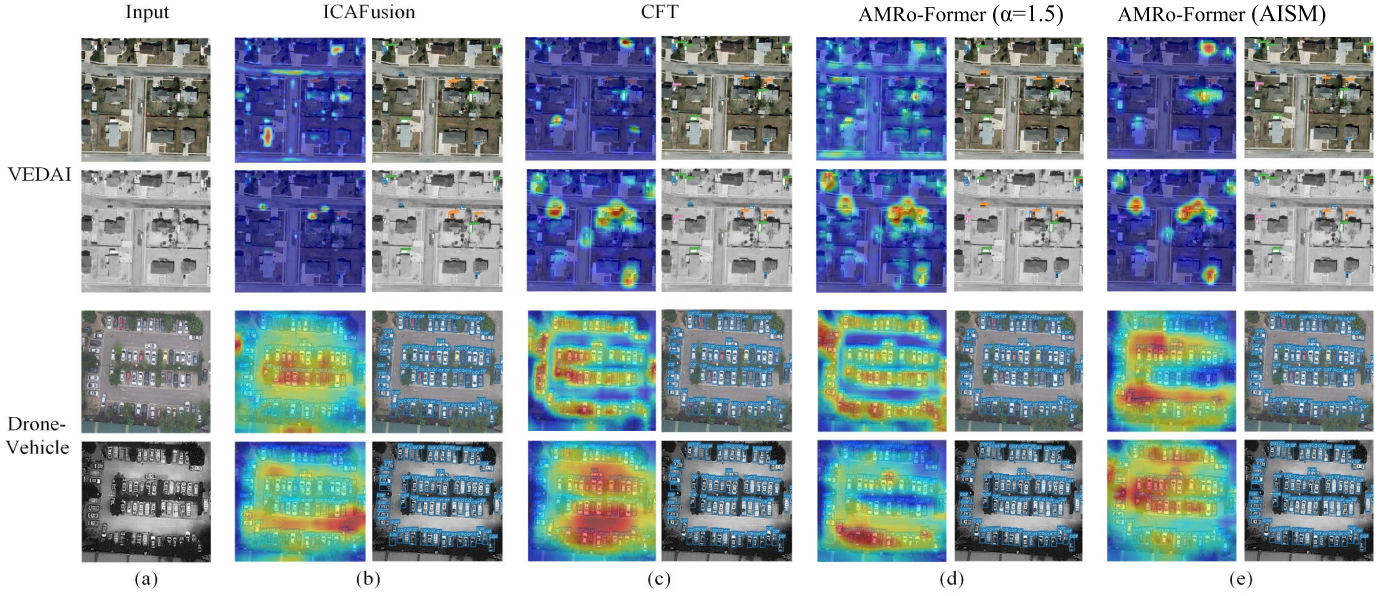


Fig. 7. Feature map visualizations of our method compared to SOTA methods on the VEDAI and drone-vehicle datasets. The brighter colors indicate regions that the model pays more attention to (a) input, (b) ICAFusion, (c) CFT, (d) our method, which sets intermodal spacing to 1.5, and (e) our method, which uses AISM.

the widely used modality type embedding design. The results indicate that when using 2-D RoPE, object detection performance showed a noticeable improvement compared to the standard CA module, which highlights that enhancing spatial awareness through 2-D positional encoding significantly boosts

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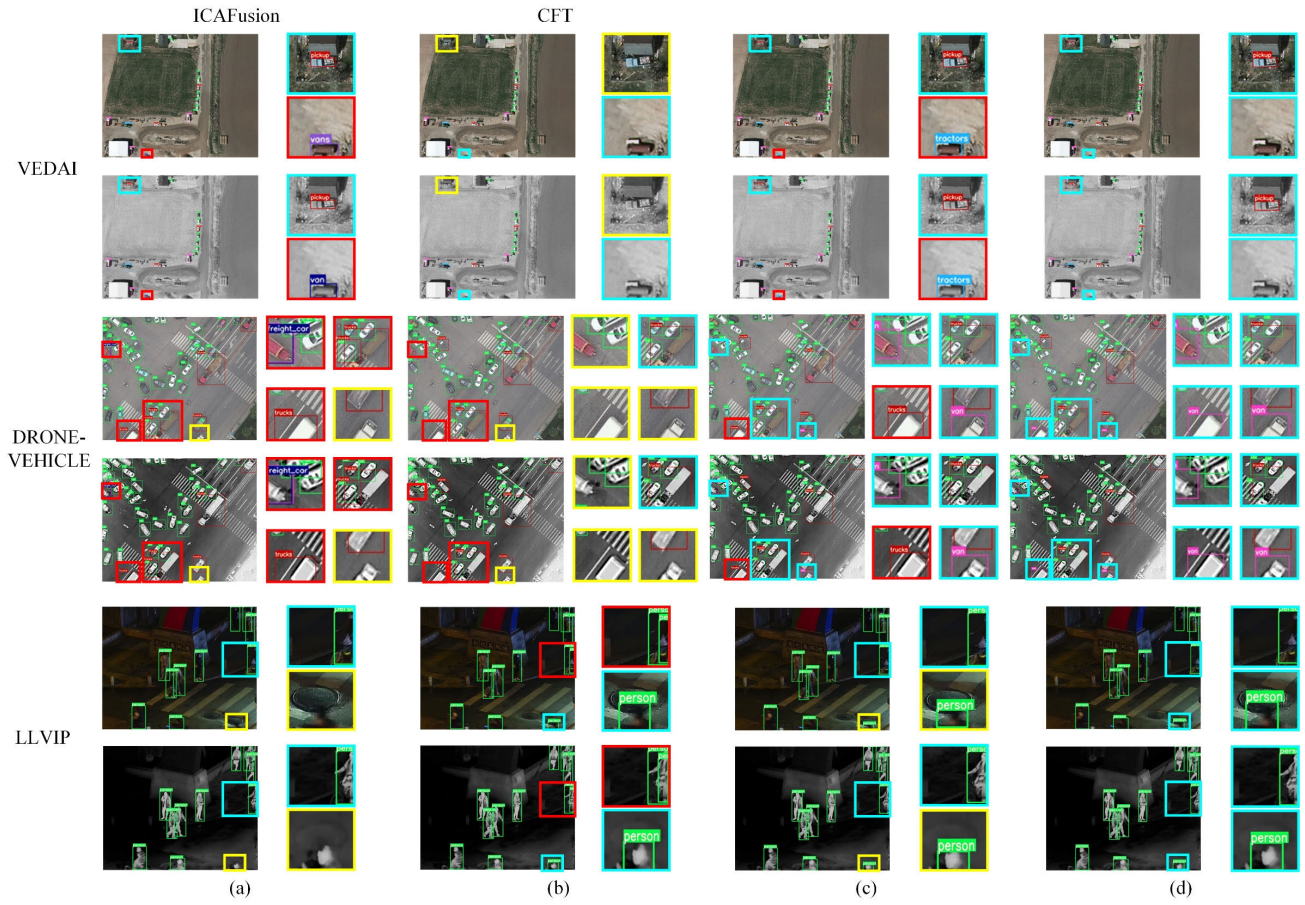


Fig. 8. Detection results of the SOTA algorithms compared to our proposed method on the VEDAI, drone-vehicle, and LLVIP datasets. We have highlighted the regions where the detection results differ between the algorithms. The blue box represents correct detection results, the red box indicates incorrect detection results, and the yellow box denotes missed detection results (a) ICAFusion, (b) CFT, (c) our method, which sets intermodal spacing to 1.5, and (d) our method, which uses AISM.

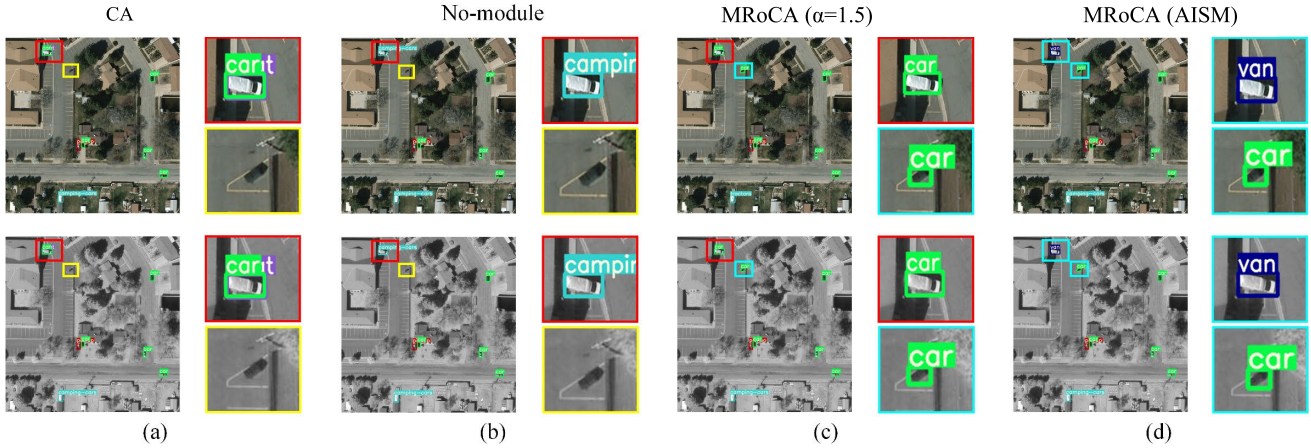


Fig. 9. Detection results obtained by replacing the MRoCA and AISM location in the dual-modal object detection architecture with a Transformer and no-module. The test was conducted on the VEDAI dataset. We have highlighted the regions where the detection results differ between the algorithms. The blue bbox represents correct detection results, the red bbox indicates incorrect detection results, and the yellow bbox denotes missed detection results (a) CA module, (b) no-module, (c) MRoCA, and (d) AMRo-Former.

both training efficiency and overall performance. On the other hand, MRoPE outperforms 2-D RoPE+APE by 0.0718, 0.1438, and 0.0688 in mAP50, mAP75, and mAP, respectively. This demonstrates the necessity of RPE in the modality position dimension.

3) *Accuracy–Efficiency Tradeoff Under Alternative Fusion Instantiations:* We adopt the standard token-wise CA formulation as a controlled baseline, in order to validate the proposed MRoPE without confounding factors introduced by specific efficient attention designs.

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TABLE IV
PER-CLASS DETECTION PERFORMANCE ON THE DRONE-VEHICLE DATASET

Method	Modal	car	freight car	truck	bus	van	mAP50	Param (m)	GFLOPs	FPS
YOLOv8L	IR	0.966	0.634	0.789	0.969	0.663	0.804	43.5	164.8	18
YOLOv8L	RGB	0.955	0.651	0.780	0.965	0.670	0.804	43.5	164.8	18
YOLOv9C	IR	0.951	0.673	0.806	0.973	0.669	0.815	50.5	236.7	16
YOLOv9C	RGB	0.965	0.688	0.802	0.974	0.676	0.821	50.5	236.7	16
GHOST	IR	0.949	0.566	0.692	0.944	0.544	0.736	-	-	20
GHOST	RGB	0.957	0.572	0.723	0.950	0.558	0.752	-	-	20
Gliding Vertex	RGB+TIR	0.802	0.450	0.531	0.779	0.450	0.602	41.1	107.2	16
C^2Former	RGB+TIR	0.885	0.555	0.627	0.808	0.584	0.692	101.0	258.3	21
SUPER-YOLO	RGB+TIR	0.924	0.497	0.703	0.947	0.581	0.724	7.0	81.5	23
ICAFusion	RGB+TIR	0.970	**0.798**	0.837	0.962	0.648	0.843	120.2	180.0	16
LF-MDet	RGB+TIR	0.822	0.596	0.736	0.866	0.570	0.718	38.7	77.7	-
DUAL-YOLOv9C	RGB+TIR	0.967	0.664	0.746	0.959	0.631	0.793	50.7	237.4	16
CFT	RGB+TIR	0.981	0.742	0.834	0.970	0.716	0.849	206.0	224.4	15
DAMSDet	RGB+TIR	0.946	0.703	0.812	0.934	0.692	0.817	79.06	83.0	-
MiPA	RGB+TIR	0.968	0.685	0.764	0.963	0.642	0.804	217.9	139.6	-
OAFA	RGB+TIR	0.975	0.732	0.815	0.968	0.672	0.799	-	-	-
AMRo-Former ($\alpha = 1.5$)	RGB+TIR	0.981	0.742	0.838	0.977	**0.723**	0.852	264.8	224.8	15
AMRo-Former (AISM)	RGB+TIR	**0.984**	0.740	**0.843**	**0.979**	0.720	**0.853**	265.1	226.5	15

TABLE V
BASELINE ARCHITECTURES AND IMPLEMENTATION DETAILS

Setting name	PE method	Modality encoding	α
MRoCA+AISM	MRoPE	z-axis rotation	AISM
MRoCA ($\alpha = 1.5$)	MRoPE	z-axis rotation	1.5
2D RoPE + APE	2D RoPE/APE	Type embedding	/
2D RoPE	2D RoPE	/	/
CA	APE	/	/

TABLE VI
ABLATIVE EXPERIMENTS FOR AMRO-FORMER IN VEDAI

Setting name	mAP50	mAP75	mAP
MRoCA(AISM)	**0.8146**	**0.6153**	**0.5219**
MRoCA ($\alpha = 1.5$)	0.7827	0.5897	0.4916
CA	0.6468	0.4130	0.3836

Nevertheless, considering the practical concern on model complexity, we further compare several alternative instantiations of the fusion block that aim to reduce computation, while keeping the rest of the architecture and training protocol unchanged.

Specifically, we evaluate: 1) the standard token-wise CA, whose attention matrix is defined over spatial tokens (WH $\times$ WH); 2) a lightweight DWConv-based fusion (MRoCA-DW), which replaces the FFN with a depth-wise convolutional block in Transformer [60]; and 3) a multi-Dconv head transposed attention (MDTA) variant (CA-MDTA/MRoCA-MDTA), where attention is computed along channels (C $\times$ C) rather than spatial tokens [61]. The results are summarized in Table VIII (mAP50, Params, GFLOPs). As shown in Table VIII, the DWConv-based fusion achieves a favorable accuracy–efficiency tradeoff: it substantially reduces Params/GFLOPs while retaining competitive mAP50, suggesting that lightweight fusion is a promising direction for deployment-oriented optimization (although it is not the main focus of this work). In contrast, the MDTA-based variants exhibit a clear accuracy drop despite their lower computational cost. We attribute this degradation to the fact that MDTA factorizes attention along channels and thus weakens explicit token-wise spatial correspondence (i.e., pixel-level interactions), which is central to our 3-D spatial–modality

TABLE VII
ABLATIVE EXPERIMENTS FOR PE METHOD IN VEDAI

Setting name	mAP50	mAP75	mAP
MRoCA ($\alpha = 1.5$)	**0.7827**	**0.5897**	**0.4916**
2D RoPE + APE	0.7109	0.4459	0.4228
2D RoPE	0.6600	0.4379	0.4008
CA	0.6468	0.4130	0.3836

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TABLE VIII
ACCURACY–EFFICIENCY COMPARISON UNDER DIFFERENT ARCHITECTURAL INSTANTIATIONS

Setting name	mAP50	Params (M)	GFLOPs (G)
CA	0.6468	264.5	224.8
CA-MDTA	0.5729	264.5	201.4
MRoCA-MDTA	0.6018	264.8	206.1
MRoCA-DW	0.7464	204.5	209.8
MRoCA(α = 1.5)	0.7827	264.8	224.8

TABLE IX
SETTING OF INTERMODAL SPACING IN VEDAI

ΔZ	mAP50	mAP75	mAP
0.5	0.7700	0.4422	0.4319
1	0.7692	0.5637	0.4800
1.28	0.7918	0.5834	0.4901
1.3	0.8013	0.6052	0.5072
1.5	0.7827	0.5897	0.4916
2	0.7419	0.5351	0.4585
α	**0.8146**	**0.6153**	**0.5219**

positional modeling that relies on preserving cross-modal spatial geometry.

Overall, this ablation indicates that reducing model capacity alone does not necessarily harm performance if the fusion mechanism still preserves token-wise spatial correspondence, whereas efficiency modifications that remove explicit spatial token interactions may undermine the geometric premise required by MRoPE.

V. DISCUSSION

A. Setting of Intermodal Spacing

Note that the intermodal spacing implicitly influences the cross-modal interaction of multimodal features during the computation of the attention mechanism, and there exists an optimal solution to achieve the best model performance. To explore the optimal solution settings of intermodal spacing, we defined two intermodal spacing setting methods: fixed setting and adaptive setting. In the fixed setting, the intermodal spacing value is manually set, while in the adaptive setting, AISM is used to automatically determine the intermodal spacing. The fixed setting typically requires a significant amount of manual trial effort and relies on empirical settings. In contrast, the adaptive setting greatly reduces the cost of manually setting a fixed multimodal spacing with slightly increased parameters.

To investigate the model performance under both fixed and adaptive settings, we conducted thorough comparative experiments. For clarity, we denote the intermodal spacing by the symbol $\Delta\mathcal{Z} = \{\Delta z_1, \Delta z_2, \Delta z_3, \Delta z_4, \Delta z_5\}$, where each Δz_i represents the intermodal spacing for each stage. In the fixed setting experiment, we fixed the intermodal spacing $\Delta\mathcal{Z}$ (consistently across all stages) to 0.5, 1, 1.28, 1.3, 1.5, and 2.

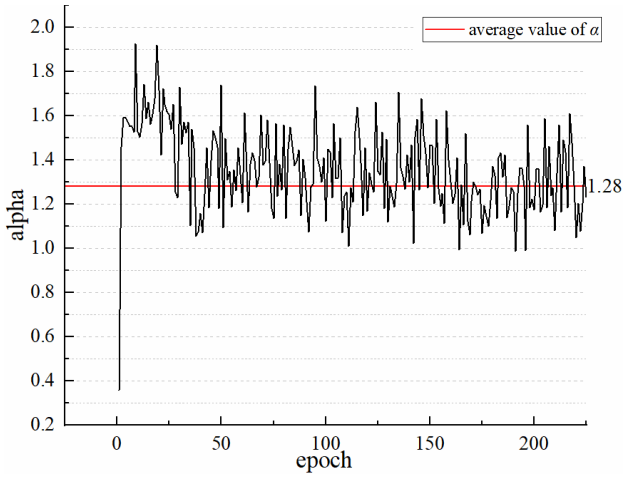


Fig. 10. Variation and average value of adaptive intermodal spacing (α) for S1 during training.

In the adaptive setting experiment, we incorporated the AISM module, and the intermodal spacing is adaptively adjusted. Experimental results demonstrate that different intermodal spacings $\Delta\mathcal{Z}$ significantly impact the model's performance on the VEDAI dataset, as shown in Table IX. We also consolidated the results related to positional encoding from our previous ablation experiments into Table IX for systematic comparison. The experimental results indicate that the adaptive setting (with AISM) achieved the best performance in terms of mAP50, mAP75, and mAP.

We collect a set of alpha values in the first stage, and after statistical analysis, as shown in Fig. 10. After the first few epochs, the predicted α converged to a stable operating range with an average around 1.28 in the later training epochs, which partly explains why fixing ΔZ near this value yields relatively strong performance. We notice that α exhibits non-negligible variations across different stages and samples, indicating that AISM performs stage and sample-dependent adjustment within a stable operating regime rather than drifting arbitrarily. More critically, no matter which fixed value is chosen, its performance still lags behind the adaptive spacing approach (see Table X). We attribute this to the fact that MRoCA modules at different network stages operate on multimodal features with different semantics and scales, and thus their optimal intermodal spacings are not identical. A fixed Δz forces the model to passively adapt to a single spacing, whereas an explicit stage or data adaptive adjustment (as in AISM) is more effective. Moreover, enabling each stage to adjust its intermodal spacing adaptively also avoids the need for extensive grid search over this hyperparameter.

To further analyze the model's performance, we extracted the feature maps of the model and visualized them, as shown in Fig. 11. From the heatmaps, we observe that the adaptive approach effectively exploits the complementary advantages of both modalities. Both the experimental results and feature maps confirm the value of adaptive multimodal spacing. It not only reduces the computational cost of manually searching for optimal solutions but also allows the model to dynamically adjust its optimal settings based on different features, thereby improving its performance.

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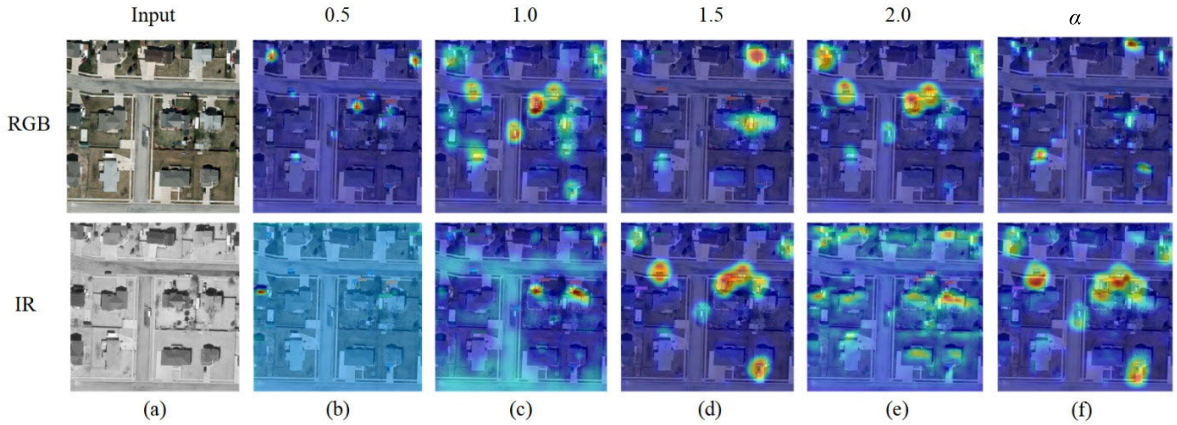


Fig. 11. Feature map visualizations of our method on the VEDAI dataset without using AISM, with fixed settings and varying intermodal spacing. Brighter colors indicate the regions that the model focuses on more (a) input, (b) $\Delta\mathcal{Z} = 0.5$, (c) $\Delta\mathcal{Z} = 1.0$, (d) $\Delta\mathcal{Z} = 1.5$, (e) $\Delta\mathcal{Z} = 2.0$, and (f) $\Delta\mathcal{Z} = \alpha$.

TABLE X
EXPERIMENTS FOR MODULE, ROPE, ABSOLUTE/RELATIVE, CA, AND $\Delta\mathcal{Z}$ IN VEDAI

Module	PE method	Attention mechanism	ΔZ	mAP50	mAP75	mAP	Precision	Recall
MRoCA	2D RoPE	Cross-attention	-	0.6600	0.4379	0.4008	0.6311	0.6096
MRoCA	2D RoPE+APE	Cross-attention	-	0.7109	0.4459	0.4228	0.6358	0.6970
MRoCA	MRoPE	Cross-attention	0.5	0.7700	0.4422	0.4319	0.7140	0.7291
MRoCA	MRoPE	Cross-attention	1	0.7692	0.5637	0.4800	0.7533	0.7108
MRoCA	MRoPE	Cross-attention	1.28	0.7918	0.5834	0.4901	0.7685	0.7231
MRoCA	MRoPE	Cross-attention	1.3	0.8013	0.6052	0.5072	**0.7955**	0.7387
MRoPE+SA	MRoPE	Self-attention	1.5	0.7412	0.5105	0.4530	0.6668	0.7211
MRoCA	MRoPE	Cross-attention	1.5	0.7827	0.5897	0.4916	0.7271	0.7259
MRoCA	MRoPE	Cross-attention	2	0.7419	0.5351	0.4585	0.6841	0.7125
MRoCA	MRoPE	Cross-attention	AISM	**0.8146**	**0.6153**	**0.5219**	0.7882	**0.7513**

B. Module Deployment in AMRo-Former

Note that the advantage of MRoCA lies in its ability to explicitly encode and preserve both spatial and modality-specific positional information for cross-modal features, enabling the model to better perform cross-modal interactions. In the input and shallow network layers, the dual-stream features primarily represent the two modalities separately, and at this stage, the modality-specific positional information is highly valuable. However, in deeper layers of the dual-stream backbone, features from one modality become progressively influenced (and diluted) by the other modality. At this point, the dual-stream features tend to represent more abstract, within-stream characteristics rather than the raw modality characteristics. Thus, the role of MRoCA and the necessity of deploying it at all network stages should be carefully evaluated.

We conducted tests on the MRoCA module at different network stages. This module can be configured as: MRoCA (√), CA (), or no module (×), to evaluate the performance of our design across different network stages. Here, "CA" denotes the standard cross-attention module, which is used here to update the feature of each branch [see (28)]. "No module" means that no fusion module is used at that stage, and the features in each branch are forwarded without cross-branch interaction [see (29)]. In these experiments, we did not use AISM (fixing α to 1.3) to isolate the impact of the variability introduced by AISM. The results are shown in Table XI.

The results of experiments 1 and 2 in Table XI indicate that in the shallow network S1, where the modality differences in the input data are most pronounced, MRoCA can effectively harness the modality information. By leveraging the modality and spatial positional information provided by MRoPE, the model achieves more effective cross-modal feature interaction. However, as the network deepens, experiments 2–4 show that deeper network layers dilute the modality information of features, limiting MRoPE's effectiveness. In addition, at the S2 level, the Transformer fails to efficiently extract high-order semantic information, resulting in poor performance. Nevertheless, compared to this Transformer baseline, AMRo-Former demonstrates relatively better performance, suggesting that MRoPE, although limited in the middle layers, still retains some effective performance. According to the results of experiment 3–8, we observe that in relatively deep, yet not the deepest, network layers, the inclusion of MRoPE in AMRo-Former leads to better performance compared to the CA module alone. At this point, MRoPE continues to leverage severely diluted modality information to slightly enhance cross-modal interaction.

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TABLE XI
EXPERIMENTS FOR THE DEPLOYMENT LOCATION OF AMRO-FORMER IN VEDAI. √ DENOTES THE MROCA MODULE, × DENOTES NO-MODULE BETWEEN BRANCHES, AND DENOTES THE STANDARD CA MODULE

Num	Stage1	Stage2	Stage3	Stage4	Stage5	mAP50	mAP75	mAP
1	▽	▽	▽	▽	▽	0.709	0.465	0.426
2	√	▽	▽	▽	▽	0.762	0.531	0.468
3	√	×	▽	▽	▽	0.877	<u>0.723</u>	0.579
4	√	√	▽	▽	▽	0.878	0.702	<u>0.582</u>
5	√	×	√	▽	▽	<u>0.886</u>	0.656	0.565
6	√	√	√	▽	▽	0.790	0.596	0.509
7	√	×	√	√	▽	**0.903**	**0.763**	**0.618**
8	√	√	√	√	▽	0.820	0.635	0.525
9	√	×	√	√	√	0.873	0.688	0.570

Based on the results of experiments 7 and 9, we find that in the deepest network layers, MRoCA is almost unable to utilize modality information. Based on these findings, we conclude that MRoCA is most effective in the shallow network layers but gradually loses its effectiveness as the network deepens. In addition, since MRoPE relies on the Transformer's self-attention mechanism, evaluating AMRo-Former's performance requires considering both the gains from MRoPE and the inherent feature-processing capability of the CA mechanism. From the experimental results, we argue that in the shallowest layers, the improvement brought by MRoPE outweighs the CA mechanism's limited ability to extract high-order semantic information at those layers. However, as the network deepens, the Transformer's power dominates the overall performance, and MRoPE's influence significantly diminishes. To balance model performance and complexity, we propose a general dual-stream backbone design that deploys MRoCA at stages S1 and S3–S4.

VI. CONCLUSION

This article proposes AMRo-Former for multispectral remote sensing object detection that efficiently utilizes multimodal information via positional encoding. The key idea is to model spatial and modality information in a unified 3-D positional space. To this end, we introduce MRoPE and embed it into CA to form MRoCA, enabling explicit and efficient cross-modal interaction. We further propose AISM to automatically learn an appropriate intermodal spacing along the modality axis, avoiding manual tuning. Experiments on multiple benchmarks demonstrate that AMRo-Former achieves competitive or SOTA performance, validating the effectiveness of 3-D spatial-modality positional modeling for multispectral detection.

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